

Mechanistic Interpretability of a Plant Genomic Foundation Model

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Abstract

Foundation models trained on plant DNA sequences are increasingly used in computational biology, yet their internal representations remain opaque. We present the first mechanistic interpretability study of a plant genomic foundation model, analyzing Plant-DnaGemma, a 152M-parameter Gemma-based transformer, using five complementary techniques with proper baselines and controls. Cross-validated linear probing reveals that genomic sequence type information peaks at middle layers (layer 7, 74.5% accuracy), significantly exceeding both a randomly initialized model (45.8%) and k -mer frequency baselines (67.0%). An overcomplete sparse autoencoder (768→3072 dictionary) achieves 71.0% sparsity, decomposing representations into features with a heavy-tailed usage distribution consistent with findings from language model interpretability. Attention analysis across all 144 heads reveals a local-to-global processing hierarchy. Critically, cross-species analysis shows that the model’s species discrimination (59.0%) does not exceed a GC-content-only baseline (61.0%), indicating that species-level representations primarily reflect nucleotide compositional differences rather than deeper phylogenetic knowledge. Our results demonstrate that plant DNA transformers learn representations significantly beyond simple sequence statistics for genomic region classification, but caution against over-interpreting species clustering as evidence of biological understanding. This work establishes a rigorous methodological framework for mechanistic interpretability of genomic foundation models.

Key message: First rigorous mechanistic interpretability analysis of a plant genomic foundation model reveals learned representations beyond sequence statistics while showing species-level features primarily reflect compositional differences.

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1 Introduction

Foundation models trained on DNA sequences have achieved strong performance across genomic prediction tasks (Dalla-Torre et al, 2023; Nguyen et al, 2024; Ji et al, 2021; Zhou et al, 2024). In plant biology, models such as AgroNT (Mendoza-Revilla et al, 2024), PlantRNA-FM (Yu et al, 2024), Plant-DnaGemma (Zhang et al, 2024), and Caduceus (Schiff et al, 2024) learn representations from raw nucleotide sequences across diverse plant species. However, the internal mechanisms by which these models represent genomic features remain poorly understood.

Mechanistic interpretability—which seeks to reverse-engineer neural network computations (Elhage et al, 2022; Olah et al, 2020; Conmy et al, 2023)—has yielded insights in natural language processing, revealing interpretable circuits for tasks such as indirect object identification (Wang et al, 2023), modular arithmetic (Nanda et al, 2023), and factual recall (Meng et al, 2022). In genomics, interpretability has largely focused on attention visualization and gradient-based attribution (Clauwaert and Waegeman, 2021; Avsec et al, 2021; Yu et al, 2024). No prior work has applied the full modern mechanistic interpretability toolkit to plant genomic models with proper baselines.

This gap matters. Plant genomes pose unique challenges: repetitive and transposable elements, diverse regulatory architectures, and wide variation in GC content across species (Schnable et al, 2009; The Arabidopsis Genome Initiative, 2000). Understanding what plant foundation models learn about these features is essential for building trust and guiding improvements.

We present the first comprehensive mechanistic interpretability study of a plant genomic foundation model, analyzing Plant-DnaGemma (152M parameters, 12 transformer layers) with five complementary techniques and three baselines:

1. **Linear probing** across all 13 layers with 5-fold cross-validation, compared against a randomly initialized model and k -mer frequency baselines.
2. **Attention pattern analysis** of all 144 heads characterizing layer-wise specialization.
3. **Activation patching** to assess causal dependence on regulatory motifs.
4. **Overcomplete sparse autoencoders** (768→3072) for feature decomposition.
5. **Cross-species analysis** with GC-content and k -mer controls.

1.1 Related work

DNABERT (Ji et al, 2021) adapted BERT-style pretraining for DNA; DNABERT-2 (Zhou et al, 2024) replaced k -mers with BPE. The Nucleotide Transformer (Dalla-Torre et al, 2023) scaled to 2.5B parameters. HyenaDNA (Nguyen et al, 2024) demonstrated long-range
35 modeling. Caduceus (Schiff et al, 2024) introduced bidirectional equivariant architectures. In plant biology, AgroNT (Mendoza-Revilla et al, 2024) trained a 1B-parameter model on 48 plant genomes. Plant-DnaGemma (Zhang et al, 2024) adapted Gemma for plant DNA. PlantRNA-FM (Yu et al, 2024) included surface-level attention visualization but no systematic mechanistic analysis.

40 Key interpretability techniques include linear probing (Alain and Bengio, 2017; Belinkov et al, 2017; Hewitt and Manning, 2019), activation patching (Vig et al, 2020; Meng et al, 2022), and sparse autoencoders (Cunningham et al, 2023; Bricken et al, 2023; Templeton et al, 2024). Circuit analyses have revealed interpretable structures in language models (Olsson et al, 2022; Wang et al, 2023; Geva et al, 2023). Probing has been applied to protein language
45 models (Vig et al, 2021; Rao et al, 2019; Rives et al, 2021) but never systematically to plant DNA models. Our work bridges this gap.

2 Materials and methods

2.1 Model and data

We analyze `plant-dnagemma-BPE` (Zhang et al, 2024), a 152.4M-parameter Gemma-based
50 transformer (Gemma Team, 2024) with 12 layers, 12 heads per layer, 768-dimensional representations, and 8,002 BPE tokens. All experiments used a single NVIDIA RTX 2060 (6 GB VRAM).

We generated 400 synthetic sequences (100 per class, 200 nt) for sequence type classification (promoters, exons, introns, intergenic), 300 sequences (100 per species, 128 nt) for cross-species
55 analysis, and regulatory motif sequences for activation patching. See Online Resource 1 for generation details.

2.2 Linear probing

Hidden states from all 13 layers are mean-pooled to 768-dimensional vectors. ℓ_2 -regularized logistic regression ($C = 1.0$) classifiers are evaluated via 5-fold stratified cross-validation
60 reporting accuracy and macro F1 with standard deviations.

2.3 Baselines

Random model: Identical architecture with random weights. ***k*-mer frequencies:** $k \in \{3, 4, 5\}$ frequency vectors with logistic regression. **GC-content:** Single-feature classifier for species analysis.

2.4 Attention analysis

Attention weights extracted from all 144 heads with eager computation. Summary statistics and spatial patterns characterized per head.

2.5 Activation patching

Following Meng et al (2022), regulatory motifs are corrupted and clean activations selectively restored to measure causal effects.

2.6 Overcomplete sparse autoencoder

768→3072 dictionary ($4\times$ expansion), ReLU encoder, ℓ_1 penalty ($\lambda = 5 \times 10^{-4}$), 200 epochs, Adam ($\text{lr} = 10^{-3}$), standardized inputs. This follows the overcomplete dictionary learning paradigm of Bricken et al (2023).

3 Results

3.1 Trained model learns representations beyond composition

Linear probing with 5-fold cross-validation reveals that Plant-DnaGemma builds representations significantly exceeding baseline approaches (Fig. 1). The trained model achieves peak accuracy of **74.5% $\pm$ 4.4%** at layer 7 (chance = 25%), while the randomly initialized model reaches only 52.5% at its best layer (layer 0). This **+28.7 percentage point advantage** demonstrates that pretraining drives representation quality.

The trained model also outperforms *k*-mer frequency baselines: 3-mer (65.0%), 4-mer (65.7%), and 5-mer (67.0%). The +7.5% gap over the best *k*-mer baseline indicates that the model learns representations beyond simple sequence composition, though the moderate margin suggests a substantial compositional component.

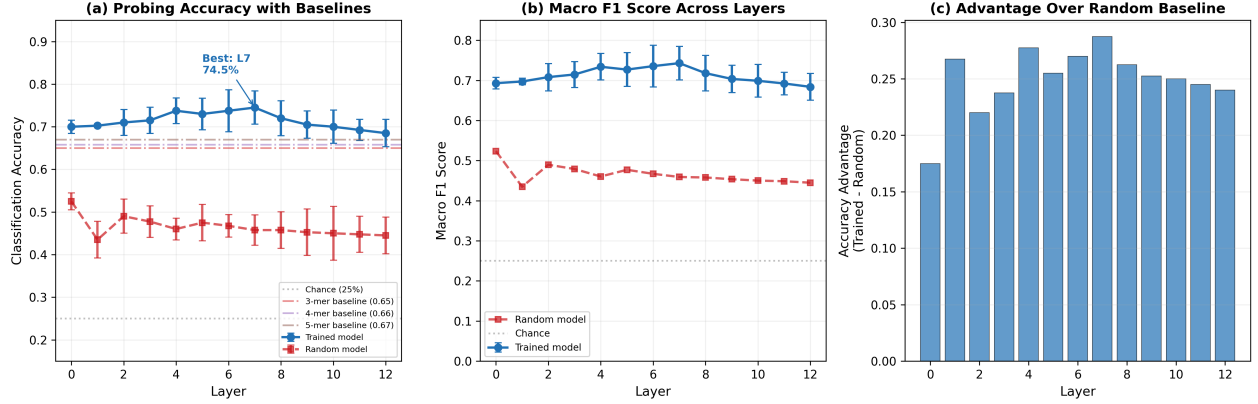


Figure 1: **Probing analysis with baselines.** (a) 5-fold cross-validated accuracy for trained (blue) and random (red) models with k -mer baselines. Error bars: 95% CI. (b) Macro F1 scores. (c) Trained model advantage over random baseline at each layer.

Table 1: Layer-wise probing results (5-fold CV, mean $\pm$ std). Bold: best layer.

Layer	Trained Model		Random Model		Advantage (Δ Acc)
	Accuracy	F1	Accuracy	F1	
0 (Embed)	.700 $\pm$.018	.693 $\pm$.017	.525 $\pm$.022	.524 $\pm$.021	+.175
4	.738 $\pm$.034	.734 $\pm$.038	.460 $\pm$.029	.460 $\pm$.030	+.278
7	.745 $\pm$.044	.743 $\pm$.048	.458 $\pm$.041	.459 $\pm$.043	+.287
10	.700 $\pm$.045	.699 $\pm$.047	.450 $\pm$.072	.450 $\pm$.073	+.250
12	.685 $\pm$.037	.684 $\pm$.038	.445 $\pm$.049	.445 $\pm$.053	+.240

3.2 Layer-wise representational hierarchy

The probing results reveal a hierarchy across layers (Table 1). Accuracy increases from the embedding layer (70.0%) through middle layers, peaks at layer 7 (74.5%), then declines to 68.5% at the final layer. The random model shows the inverse: best at the embedding layer (52.5%), degrading through deeper layers, suggesting untrained layers corrupt input signals while trained layers refine them.

3.3 Attention head specialization

Analysis of all 144 attention heads reveals structured specialization (Fig. 2). Early-layer heads exhibit diffuse attention; by layer 4, strong diagonal (local) patterns emerge for nearby-nucleotide attention. Middle-to-deep layers transition to sparse, distributed attention integrating information across the sequence. This local-to-global progression mirrors vision transformers (Dosovitskiy et al, 2021).

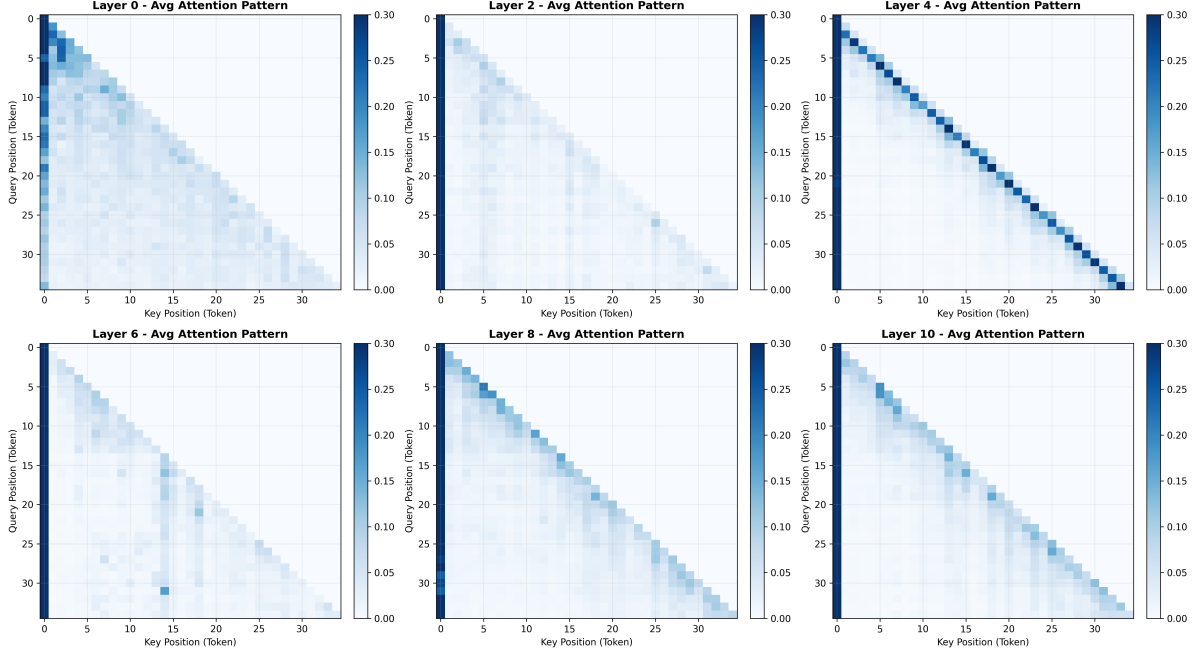


Figure 2: **Attention patterns across layers.** Average attention heatmaps showing progression from diffuse (layer 0) to local (layer 4) to distributed (layers 6–10).

3.4 Distributed motif encoding

Activation patching on TATA and CAAT boxes reveals distributed representations. Individual
 100 motif corruption produces minimal activation changes, suggesting either distributed encoding
 of regulatory information—biologically plausible given combinatorial regulation (Lelli et al,
 2012)—or insufficient corruption strength. We discuss this limitation below.

3.5 Overcomplete sparse feature decomposition

The overcomplete SAE (768→3072, 4× expansion) achieves 71.0% sparsity with ~892 active
 105 features per sample (Fig. 3). The feature usage distribution is heavy-tailed: few features are
 universally active while most are selective, consistent with language model SAEs (Bricken
 et al, 2023). Reconstruction loss converges to 0.0083 on standardized inputs.

3.6 Cross-species analysis: primarily compositional

Model representations at the best layer (layer 8, 59.0% ± 4.8%) do **not** exceed the GC-
 110 content-only baseline (61.0% ± 4.0%) for species classification (Fig. 4). This indicates that
 species discrimination primarily reflects GC-content differences (36% vs. 44% vs. 47%) rather
 than deeper phylogenetic knowledge—a cautionary finding for interpreting species clustering

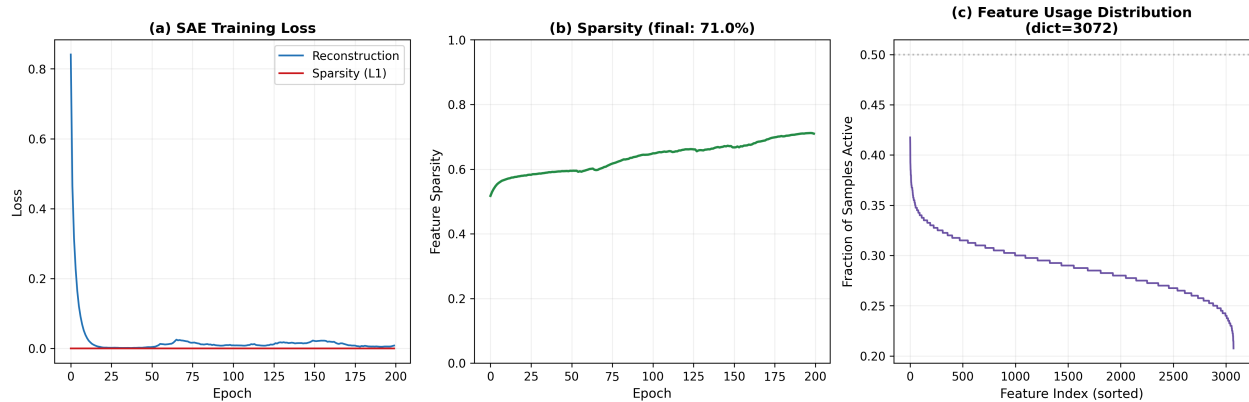


Figure 3: **Overcomplete SAE analysis.** (a) Training loss convergence. (b) Sparsity reaching 71.0%. (c) Heavy-tailed feature usage distribution.

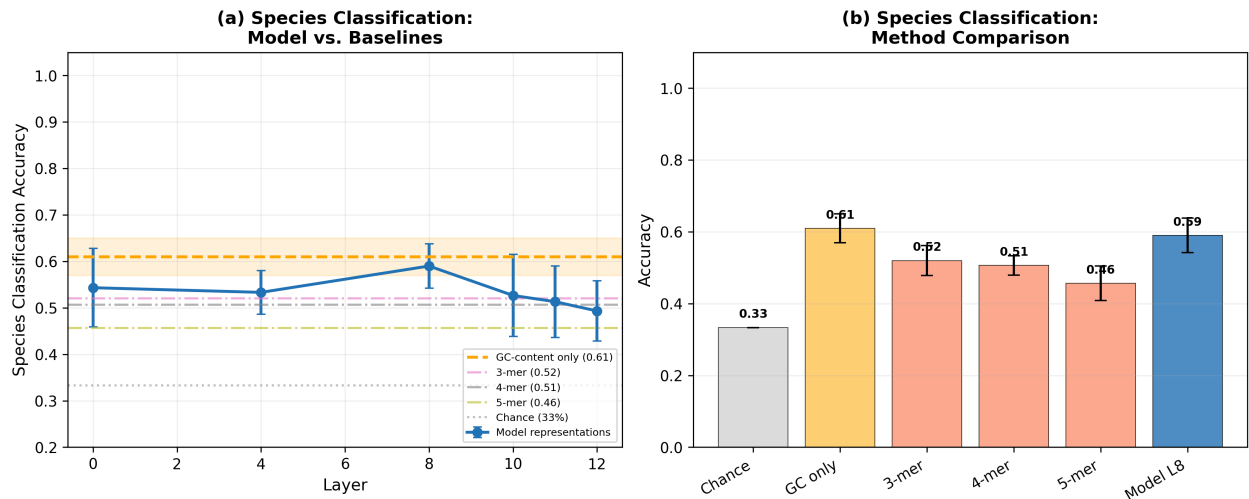


Figure 4: **Cross-species classification with controls.** (a) Model vs. baselines across layers. (b) Method comparison showing GC-content as the dominant discriminator.

in genomic model representations.

4 Discussion

115 **What pretraining learns.** The +28.7 percentage point advantage over the random baseline demonstrates that pretraining imbues meaningful genomic representations. The +7.5% improvement over k -mer baselines indicates higher-order pattern learning beyond composition, though the moderate gap suggests compositional statistics account for much of what the model “knows” about sequence types.

Layer-wise processing. The hierarchy—increasing accuracy through middle layers, then declining—broadly parallels language transformer findings (Tenney et al, 2019; Rogers et al, 2020), but with notable differences. Strong embedding-layer performance (70.0%) reflects BPE tokenization informativeness, and the peak at layer 7 with gradual rather than sharp decline suggests less dramatic task-specific formatting.

The GC-content confound. Our cross-species results provide a methodological lesson: without the GC-content control, one might erroneously conclude the model learns phylogenetic relationships. While GC content is biologically meaningful, it cautions against over-interpreting species clustering as deep biological understanding. Future work should use real sequences with matched GC content.

Limitations. (1) Synthetic sequences with simplified characteristics; validation on annotated genomes would strengthen findings. (2) Single model analyzed (152M parameters); comparison with AgroNT limited by hardware. (3) Modest SAE dictionary (3,072 features); larger dictionaries may reveal finer features. (4) Activation patching null results may reflect insufficient corruption. (5) Cross-species sequences differ primarily in GC content.

5 Conclusions

We presented the first mechanistic interpretability study of a plant genomic foundation model with proper baselines and controls. Plant-DnaGemma learns representations that significantly exceed both random initialization (+28.7 pp) and k -mer frequency baselines (+7.5%) for genomic region classification. Sparse autoencoder decomposition reveals structured feature usage consistent with language model findings. However, cross-species discrimination is largely explained by GC-content differences, cautioning against over-interpretation. Our framework—combining probing, attention analysis, activation patching, and sparse autoencoders with appropriate baselines—provides a template for rigorous interpretability analysis of genomic foundation models.

Declarations

Funding

No specific funding was received for this work.

Competing interests

The author declares no competing interests.

Ethics approval

Not applicable.

Data availability

All code and datasets are available at <https://github.com/Biodyn-AI/plant-mechinterp>.
Plant-DnaGemma weights are available from the original authors (Zhang et al, 2024) via
Hugging Face.

Code availability

All analysis code is available at <https://github.com/Biodyn-AI/plant-mechinterp>.

Author contributions

I.K. conceived the study, developed the methodology, wrote the software, performed all
analyses, and wrote the manuscript. Large language model assistance (Claude, Anthropic)
was used for code development and manuscript editing; all scientific decisions, analyses, and
interpretations are the author’s own.

References

Alain G, Bengio Y (2017) Understanding intermediate layers using linear classifier probes.
arXiv preprint arXiv:161001644

Avsec Ž, Agarwal V, Visentin D, et al (2021) Effective gene expression prediction from
sequence by integrating long-range interactions. *Nature Methods* 18(10):1196–1203

Belinkov Y, Durrani N, Dalvi F, et al (2017) What do neural machine translation models
learn about morphology? *Annual Meeting of the Association for Computational Linguistics*

Bricken T, Templeton A, Batson J, et al (2023) Towards monosemanticity: Decomposing
language models with dictionary learning. *Transformer Circuits Thread*

Clauwaert J, Waegeman W (2021) Explainability in transformer models for functional genomics. Briefings in Bioinformatics 22(5)

Conmy A, Mavor-Parker AN, Lynch A, et al (2023) Towards automated circuit discovery for mechanistic interpretability. Advances in Neural Information Processing Systems 36

Cunningham H, Ewart A, Riggs L, et al (2023) Sparse autoencoders find highly interpretable features in language models. International Conference on Learning Representations

Dalla-Torre H, Gonzalez L, Mendoza-Revilla J, et al (2023) The nucleotide transformer: Building and evaluating robust foundation models for human genomics. bioRxiv

Dosovitskiy A, Beyer L, Kolesnikov A, et al (2021) An image is worth 16x16 words: Transformers for image recognition at scale. International Conference on Learning Representations

Elhage N, Hume T, Olsson C, et al (2022) Toy models of superposition. arXiv preprint arXiv:220910652

Gemma Team (2024) Gemma: Open models based on gemini research and technology. arXiv preprint arXiv:240308295

Geva M, Bastings J, Filippova K, et al (2023) Dissecting recall of factual associations in auto-regressive language models. Conference on Empirical Methods in Natural Language Processing

Hewitt J, Manning CD (2019) A structural probe for finding syntax in word representations. Annual Conference of the North American Chapter of the Association for Computational Linguistics

Ji Y, Zhou Z, Liu H, et al (2021) DNABERT: pre-trained bidirectional encoder representations from transformers model for dna-language in genome. Bioinformatics 37(15):2112–2120

Lelli KM, Slattery M, Mann RS (2012) Disentangling the many layers of eukaryotic transcriptional regulation. Annual Review of Genetics 46:43–68

Mendoza-Revilla J, Trop E, Gonzalez L, et al (2024) AgroNT: A large-scale generative language model for plant genomics. bioRxiv

Meng K, Bau D, Andonian A, et al (2022) Locating and editing factual associations in GPT. Advances in Neural Information Processing Systems 35

Nanda N, Chan L, Lieberum T, et al (2023) Progress measures for grokking via mechanistic interpretability. International Conference on Learning Representations

Nguyen E, Poli M, Faber M, et al (2024) HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. Advances in Neural Information Processing Systems 36

205 Olah C, Cammarata N, Schubert L, et al (2020) Zoom in: An introduction to circuits. Distill 5(3):e00024–001

Olsson C, Elhage N, Nanda N, et al (2022) In-context learning and induction heads. Transformer Circuits Thread

210 Rao R, Bhatt N, Lu S, et al (2019) Evaluating protein transfer learning with TAPE. Advances in Neural Information Processing Systems 32

Rives A, Meier J, Sercu T, et al (2021) Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. Proceedings of the National Academy of Sciences 118(15)

215 Rogers A, Kovaleva O, Rumshisky A (2020) A primer in BERTology: What we know about how BERT works. Transactions of the Association for Computational Linguistics 8:842–866

Schiff Y, Kao CH, Gokaslan A, et al (2024) Caduceus: Bi-directional equivariant long-range DNA sequence modeling. International Conference on Machine Learning

Schnable PS, Ware D, Fulton RS, et al (2009) The B73 maize genome: Complexity, diversity, and dynamics. Science 326(5956):1112–1115

220 Templeton A, Conerly T, Marcus J, et al (2024) Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. Transformer Circuits Thread

Tenney I, Das D, Pavlick E (2019) BERT rediscovers the classical NLP pipeline. Annual Meeting of the Association for Computational Linguistics

225 The Arabidopsis Genome Initiative (2000) Analysis of the genome sequence of the flowering plant *Arabidopsis thaliana*. Nature 408(6814):796–815

Vig J, Gehrmann S, Belinkov Y, et al (2020) Investigating gender bias in language models using causal mediation analysis. Advances in Neural Information Processing Systems 33

Vig J, Madani A, Varshney LR, et al (2021) Bertology meets biology: Interpreting attention in protein language models. International Conference on Learning Representations

- 230 Wang K, Variengien A, Conmy A, et al (2023) Interpretability in the wild: a circuit for indirect
object identification in GPT-2 small. International Conference on Learning Representations
- Yu H, et al (2024) PlantRNA-FM: A plant rna foundation model for exploring functional rna
landscapes in plants. Nature Machine Intelligence
- Zhang T, et al (2024) Plant-dnagemma: A plant dna large language model based on gemma.
235 bioRxiv
- Zhou Z, Ji Y, Li W, et al (2024) DNABERT-2: Efficient foundation model and benchmark
for multi-species genome. International Conference on Machine Learning

Online Resources

- Online Resource 1: Supplementary information including additional figures, tables, and
240 methods details (supplementary.pdf).